

**Design and Evaluation of a QR Code-Based Web Evaluation System for
Extension Project Feedback Management in Higher Education Institutions**

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Table of Contents

	Page
Title Page	1
CHAPTER 1 – THE PROBLEM AND ITS BACKGROUND	
Introduction	1
Objectives of the Study	4
Scope and Limitation	6
Significance of the Study	7
Review of Related Literatures and Studies	8
Definition of Terms	12
CHAPTER 2 – TECHNICAL BACKGROUND	
Conceptual Framework	14
System Architecture	15
Context Diagram	17
Data Flow Diagram	18
Level 0 DFD.....	19
Level 1 DFD.....	20
Use Case Diagram	21
Entity Relationship Diagram	22
Database Schema	26
GANTT chart	28
CHAPTER – METHODOLOGY	
Research Design	30
Research Participants	38
Data Gathering Procedure	38
Research Instrument	39
Statistical Treatment	40
References.....	50

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

Introduction

In today's digital age, technological advancements have changed the way people work and manage information. Many industries, including education and community development, have started using modern tools and systems to make processes faster, more accurate, and easier to manage. Automation has become especially important, replacing time-consuming manual tasks with efficient solutions that save time and reduce mistakes. This shift has helped organizations stay organized and productive while keeping up with today's fast-paced environment.

Despite these advancements, many extension offices, including the Extension Office at Isabela State University Ilagan Campus, continue to rely on outdated manual processes for collecting evaluation feedback from partner communities. These manual processes present significant challenges, including disorganized records, delayed feedback collection, and inefficient data analysis. Evaluation forms often get misplaced, lost, or submitted late, leading to incomplete data and unreliable reports. This gap in feedback management compromises the ability of the extension office to measure project impact, make data-driven decisions, and comply with reporting requirements for funding and accreditation.

To address these issues, this study proposes a QR Code–Based Web Evaluation System tailored to the needs of the ISU Ilagan Campus Extension Office. This system aims to digitize the manual process of collecting post-project evaluation feedback. By generating unique QR codes for individual extension projects and hosting mobile-responsive evaluation forms, the system ensures that feedback

collection is streamlined, response rates are improved, and data is automatically organized. The system offers several benefits, including reduced operational expenses by minimizing printing costs, increased efficiency in compiling and analyzing feedback, and the adoption of paperless transactions for sustainable and environmentally friendly operations. Additionally, it reduces the workload of extension coordinators and staff, making evaluation management more convenient and less stressful. The cost efficiency achieved through streamlined processes further underscores the system's value.

Several studies have explored the implementation of web-based systems for evaluation and feedback management. For instance, a study titled "Web-based Faculty Development Management System" developed a database responsive to the professional growth of faculty members, facilitating organized storage and easy retrieval of faculty profiles (Bajao et al., 2022). Similarly, research on "Faculty E-Portfolio for Professional Credentialing" assessed the acceptability of a developed e-portfolio system, highlighting its potential to streamline record management and support accreditation processes (Abojon et al., 2022). These studies demonstrate that implementing web-based systems can enhance efficiency, reduce errors, and improve compliance with institutional standards. Moreover, the use of QR codes for evaluation has been shown to increase participant engagement and simplify access to digital forms. Bagulaya-Abogaa (2023) developed a smartphone-integrated academic performance evaluation tool that replaced traditional paper-based assessment methods, demonstrating improved efficiency and accessibility. Raymundo, DiVittorio, and Heiman (2025) implemented QR codes for curriculum evaluation in medical education, reporting a 40% increase in response rates and significantly higher-quality qualitative comments. Smith, Dever, Hopkins, and Murphy (2025) examined QR code-

based survey recruitment across multiple research contexts, finding that QR code recruitment yielded response rates 25-35% higher than traditional methods with faster completion times and lower data entry errors. These studies confirm that QR codes are a practical solution for improving data collection efficiency across various educational and research settings.

However, existing QR code-based evaluation systems have primarily been designed for classroom, faculty, or general survey environments. What they fail to provide is an integrated, project-specific evaluation management platform tailored explicitly for university extension offices—one that combines dynamic QR code generation per project, mobile-responsive feedback collection, automated data validation, and real-time administrative reporting within a single system. Most current solutions focus on isolated aspects of feedback collection rather than offering a comprehensive workflow that addresses the unique needs of extension programs, including project-specific form linking, response tracking across multiple community partners, and automated report generation for accreditation compliance. Thus, this study addresses this gap by developing a QR code-based web evaluation system specifically designed for extension project feedback management, automated reporting, and real-time analytics at the Isabela State University Ilagan Campus Extension Office.

Moreover, this system has the potential to be adopted by other extension offices within the Isabela State University system in the future, extending its benefits beyond the Ilagan Campus. By addressing current challenges and introducing innovative solutions, this system ensures a more organized, transparent, and effective approach to managing extension project evaluations and related reporting tasks.

Furthermore, this study aligns with Sustainable Development Goal (SDG) 4 (Quality Education), which focuses on providing quality education and promoting lifelong learning. By introducing a QR Code–Based Web Evaluation System, this research aims to improve how educational institutions assess and improve their extension programs. Additionally, the system supports SDG 9 (Industry, Innovation, and Infrastructure) by promoting innovation through the adoption of modern technologies in community development work. Furthermore, it contributes to SDG 16 (Peace, Justice, and Strong Institutions) by enhancing transparency, accountability, and efficiency in institutional processes. Together, these alignments ensure that the system not only improves extension project evaluation but also supports global efforts to create more equitable, innovative, and effective educational institutions.

In conclusion, the QR Code–Based Web Evaluation System for the Extension Office is not only a response to the current challenges faced by the ISU Ilagan Campus Extension Office but also a forward-looking solution that promotes efficiency, sustainability, and innovation. By integrating automation and technology into evaluation processes, this system ensures a streamlined, organized, and cost-effective approach to feedback management while supporting global efforts to improve educational institutions and their community engagement operations.

Objective of the Study

The study aims to develop a QR Code–Based Web Evaluation System for Extension Office Projects at Isabela State University Ilagan Campus that is efficient, accessible, accurate, and reliable for extension personnel, project leaders, and community beneficiaries.

1. To determine the processes involved in the existing manual evaluation of extension projects.
2. To determine the problems encountered in the existing processes, such as:
 - 2.1 Collection of evaluation forms;
 - 2.2 Consolidation of data; and
 - 2.3 Generation of reports.
3. To develop a system that includes the following modules:
 - 3.1 User Account Management;
 - 3.2 Project Management;
 - 3.3 Dynamic QR Code Generator;
 - 3.4 Mobile-Responsive Evaluation Form;
 - 3.5 Admin Dashboard;
 - 3.6 Automated Data Validation;
 - 3.7 Response Management; and
 - 3.8 Reports Generation.
4. To test the developed system in terms of:
 - 4.1 Alpha testing; and
 - 4.2 Beta testing.
5. To evaluate the software quality and user acceptance of the proposed system using:
 - 5.1 ISO/IEC 25010 in terms of:
 - 5.2 Functional Suitability;
 - 5.3 Reliability;
 - 5.4 Usability;
 - 5.5 Performance Efficiency;

5.6 Security; and

5.7 Maintainability.

5.8 Technology Acceptance Model (TAM) in terms of:

5.9 Perceived Usefulness; and

5.10 Perceived Ease of Use.

5.11 Unified Theory of Acceptance and Use of Technology (UTAUT) in terms of:

5.12 Performance Expectancy;

5.13 Effort Expectancy;

5.14 Social Influence; and

5.15 Facilitation Conditions.

Scope and Limitation

The study focuses on the Extension Office of Isabela State University Ilagan Campus. The system is designed for three user groups: extension personnel (who create projects and generate QR codes), beneficiaries (who scan QR codes and submit feedback), and administrators (who manage the system). System functions include project management, dynamic QR code generation, mobile-responsive evaluation forms, an admin dashboard with real-time reporting, automated data validation, and secure response storage. The evaluation covers functional suitability, reliability, usability, performance efficiency, security, maintainability, and user acceptance using ISO/IEC 25010, TAM, and UTAUT. Technologically, the system uses static QR codes, requires internet connectivity, and does not integrate with other university systems.

The study is limited to the Extension Office of Isabela State University Ilagan Campus and does not extend to other campuses or external organizations. The

system does not support offline QR code scanning or offline form submission, requiring beneficiaries to have internet connectivity to provide feedback. The study does not evaluate long-term adoption or sustained usage of the system after the pilot deployment period. Additionally, the system does not integrate with existing university information systems such as the Student Information System (SIS), Learning Management System (LMS), or human resource and financial systems. QR codes generated by the system are static and cannot be updated after generation, and the study focuses exclusively on post-project evaluation, excluding pre-project needs assessments or long-term impact evaluations.

Significance of the Study

The findings and output of this study will benefit several groups. Extension Office personnel of Isabela State University Ilagan Campus will benefit from faster data collection, automated report generation, and improved monitoring of extension projects, reducing manual encoding tasks. Project participants and community beneficiaries can easily access and complete evaluation forms by scanning a QR code, making the feedback process more convenient, accessible, and less time-consuming. Extension coordinators and project leaders will receive accurate, real-time evaluation results to assess project effectiveness, identify areas for improvement, and make evidence-based decisions for future implementations. Finally, future researchers and developers will find a documented framework and reference for developing similar QR code-based evaluation systems in other state universities and colleges (SUCs) or government extension offices.

Review of Related Literature and Studies

QR codes have emerged as an effective tool for improving access to digital evaluation forms across various educational and research contexts. Bagulaya-Abogaa (2023) developed a smartphone-integrated academic performance evaluation tool designed to replace traditional paper-based assessment methods. The system used QR code scanner support to allow students and evaluators to instantly access evaluation forms by scanning a QR code displayed on posters, handouts, or digital screens. The study demonstrated that QR code integration significantly reduced the time between project implementation and feedback collection. However, the system was limited to academic performance tracking and did not include project-specific reporting or multi-project management capabilities. This limitation is relevant because extension offices typically manage multiple projects simultaneously, requiring a system that can organize and report on evaluations at both the individual project and program levels.

Similarly, Raymundo, DiVittorio, and Heiman (2025) implemented a QR code intervention for curriculum evaluation in medical education. The system replaced traditional paper-based course evaluation forms with QR codes printed on course syllabi, classroom posters, and email signatures. Students scanned QR codes using their smartphones and completed anonymous web-based evaluation forms. Post-implementation results showed a 40% increase in response rates and significantly higher-quality qualitative comments. While these findings confirm that QR codes enhance both the quantity and depth of evaluation data, the study focused exclusively on course evaluations within a structured academic setting. It did not address the unique needs of community-based extension programs, where participants may have varying levels of digital literacy and access to technology.

Smith, Dever, Hopkins, and Murphy (2025) examined QR code-based survey recruitment across multiple research contexts. The study deployed QR codes on printed invitations, posters, and digital screens, linking participants to web-based surveys. Results from over 5,000 participants across three separate studies showed that QR code recruitment yielded response rates 25-35% higher than traditional methods, with significantly faster completion times and lower data entry errors. The study concluded that QR codes are a highly effective tool for improving survey data collection efficiency. However, the study focused on general survey recruitment rather than structured project evaluation workflows that require authentication, project-specific forms, and administrative reporting.

Mobile-Responsive Feedback Platforms

The effectiveness of QR code-based evaluation systems depends heavily on the accessibility and usability of the feedback interface on mobile devices. Bagulaya-Abogaa (2023) addressed this by building the system with mobile-responsive web technologies, eliminating the need for dedicated mobile applications and enabling users to complete evaluations directly from their smartphone browsers. This design choice is particularly important for extension programs serving diverse community partners who may not have the ability or willingness to download specialized applications. However, the study did not evaluate the system's performance across different mobile devices, operating systems, or network conditions, leaving questions about its reliability in varied real-world settings.

Raymundo et al. (2025) similarly optimized their evaluation forms for mobile devices, recognizing that students primarily access digital content through smartphones. The study reported that the mobile-responsive design contributed to the

increased response rates by reducing friction in the feedback process. Nevertheless, the study was conducted within a university environment where students have reliable internet access and familiarity with mobile web interfaces. Extension programs often serve communities with limited connectivity and varying levels of digital literacy, a context not addressed in existing literature.

Real-Time Reporting and Analytics

A key advantage of digital evaluation systems over paper-based methods is the ability to access and analyze data in real time. Smith et al. (2025) incorporated real-time response tracking in their QR code-based survey system, allowing researchers to monitor submission rates and adjust recruitment strategies as needed. The study demonstrated that real-time data access enables more responsive decision-making during data collection. However, the system did not include automated report generation or data visualization features, requiring researchers to manually export and analyze data using external tools.

Bagulaya-Abogaa (2023) addressed automated data consolidation by designing the system to facilitate real-time submission and automated data organization. The study reported improved efficiency in academic performance tracking, but it did not include an administrative dashboard with visual analytics or customizable report generation. Extension offices require not only data collection but also the ability to generate summary reports for accreditation, funding compliance, and program improvement. Existing systems lack these integrated reporting capabilities.

Technology Acceptance in Digital Evaluation Systems

User acceptance is critical to the successful adoption of any new technology. Raymundo et al. (2025) noted that the success of their QR code intervention depended

on students' willingness to scan QR codes and complete web-based forms. The study reported high acceptance rates, but it did not apply formal technology acceptance frameworks such as TAM or UTAUT to measure perceived usefulness, perceived ease of use, or behavioral intention to use. Without such measurement, it is difficult to generalize findings to different user populations or predict adoption in new contexts.

Smith et al. (2025) similarly observed high participation rates but did not evaluate user acceptance factors that may influence willingness to engage with QR code-based systems, such as privacy concerns, trust in the technology, or facilitating conditions like internet access and device compatibility. Bagulaya-Abogaa (2023) also lacked a formal technology acceptance evaluation. None of the reviewed studies applied both ISO/IEC 25010 software quality standards and TAM/UTAUT frameworks to comprehensively evaluate a QR code-based evaluation system from both technical and user-centered perspectives.

Research Gap

Existing QR code-based evaluation systems have demonstrated improved response rates, faster data collection, and increased accessibility compared to traditional paper-based methods. However, the reviewed studies share several limitations. First, most systems are designed for classroom, faculty, or general survey environments rather than the specific context of university extension offices, which require project-specific evaluation forms, management of multiple simultaneous projects, and automated reporting for compliance purposes. Second, existing systems lack integrated reporting and analytics dashboards that allow administrators to visualize data and generate summary reports without external tools. Third, no reviewed study applied formal technology acceptance models to measure user

acceptance of QR code-based evaluation systems, leaving questions about the factors that influence adoption among extension personnel and community beneficiaries. Finally, none of the existing systems combine dynamic QR code generation per project, mobile-responsive evaluation forms, automated data validation, real-time administrative reporting, and comprehensive software quality evaluation using ISO/IEC 25010 and TAM/UTAUT within a single integrated platform. Thus, this study addresses these gaps by developing a QR code-based web evaluation system specifically tailored to the needs of university extension offices.

Definition of Terms

The following terms are defined operationally and conceptually for this study.

Agile Software Development – A methodology that emphasizes iterative development, continuous feedback, and flexibility. In this study, the Scrum framework is used to build the system in short cycles called sprints.

Extension Office Projects – Outreach initiatives, training programs, and community services implemented by the Isabela State University Ilagan Campus Extension Office. These projects are the primary subjects of evaluation in this study.

Manual Data Consolidation – The traditional process of collecting paper-based evaluation forms, manually encoding responses into digital spreadsheets, and compiling reports. This study aims to replace this process with automated digital collection.

Mobile-Responsive Interface – A web design approach that ensures evaluation forms automatically adjust to fit different screen sizes, particularly smartphones and tablets, eliminating the need for a separate mobile application.

QR Code (Quick Response Code) – A two-dimensional barcode that can be scanned using a smartphone camera to instantly access a website or digital form. In this study, QR codes are used to link participants directly to project evaluation forms.

Real-Time Data Visualization – The immediate display of submitted evaluation data through charts, graphs, and dashboards without delay. This feature enables extension personnel to make timely, evidence-based decisions.

TAM (Technology Acceptance Model) – A theory that predicts user acceptance of technology based on perceived usefulness and perceived ease of use. This study applies TAM to measure beneficiary and staff acceptance of the QR code system.

UTAUT (Unified Theory of Acceptance and Use of Technology) – A comprehensive technology acceptance model incorporating performance expectancy, effort expectancy, social influence, and facilitating conditions. This study uses UTAUT alongside TAM for user acceptance evaluation.

Web Evaluation System – A software application accessible via a web browser that allows users to complete, submit, and manage evaluation data. In this study, it refers to the custom-built platform for the ISU Ilagan Extension Office.

CHAPTER II

TECHNICAL BACKGROUND

Conceptual Framework

This study is guided by the Input-Process-Output (IPO) model, which illustrates the logical flow from planning to completion of the QR Code–Based Web Evaluation System for Extension Office Projects. The framework is visually represented in Figure 1 and described in detail below.

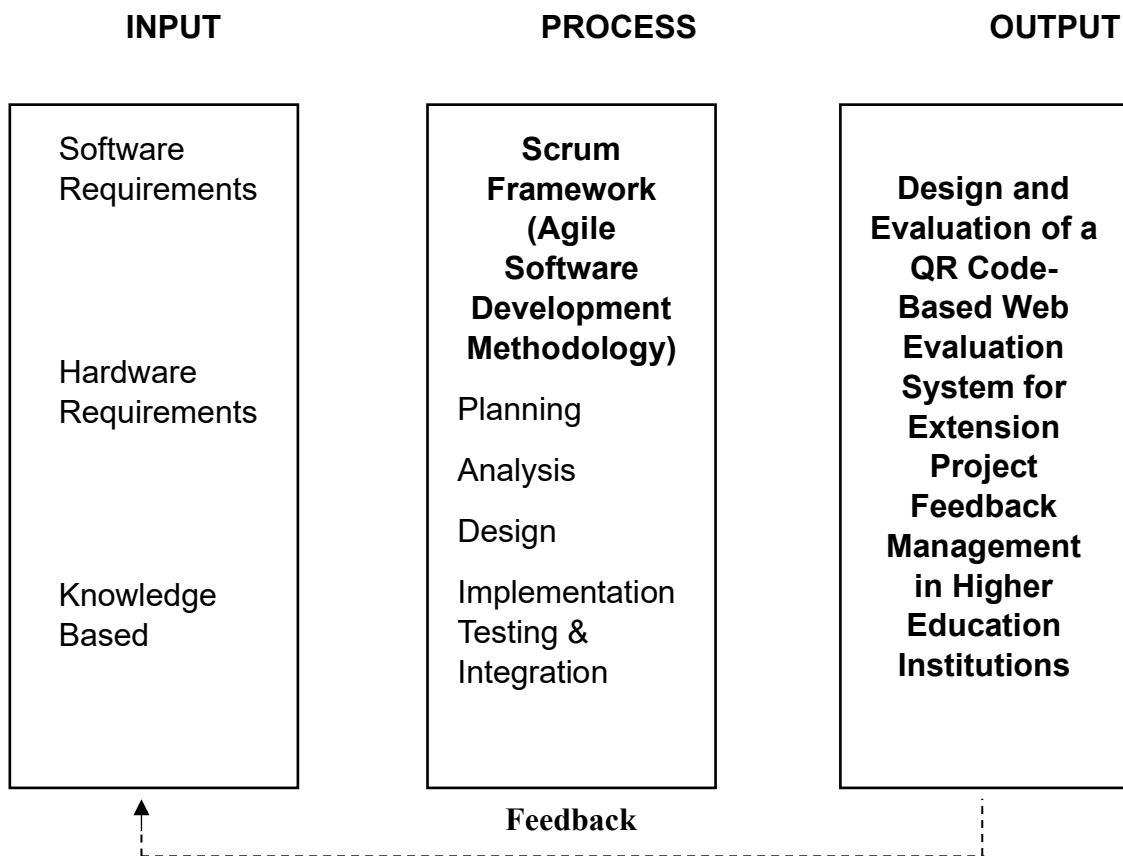


Figure 2-1. Input–Process–Output Conceptual Framework of the QR Code–Based Web Evaluation System

The Figure 2-1 illustrates the conceptual framework of the QR Code–Based Web Evaluation System for Extension Office Projects using the Input-Process-Output

(IPO) model to show how the study was conducted. The Input includes the needed knowledge in database management and web development, as well as software such as XAMPP, JavaScript, PHP, MySQL, and CSS, and hardware like desktop computers and laptops with stable internet connectivity through Wi-Fi or Ethernet. The Process follows the Scrum Framework, starting with requirements analysis to identify user needs, followed by system design, coding, testing to ensure functionality and usability, deployment for actual use, and maintenance for continuous improvement. The Output is a fully functional QR Code–Based Web Evaluation System, designed to make document submission, tracking, and archiving more organized, efficient, and reliable while helping ensure timely compliance with institutional requirements.

System Architecture

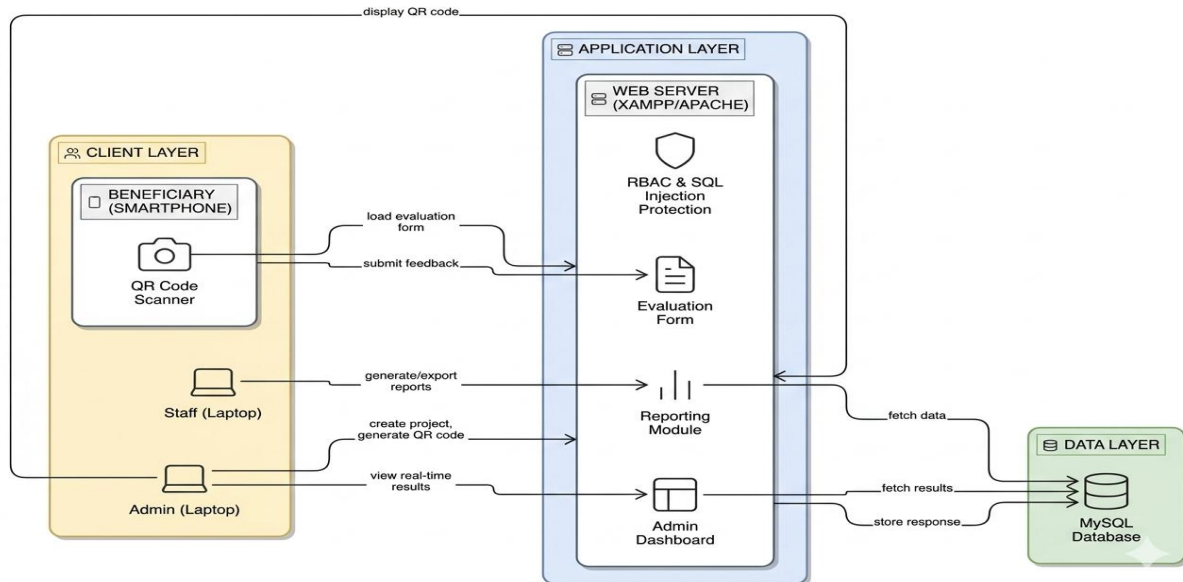


Figure 2-2. System Architecture of the QR Code–Based Web Evaluation System

The system architecture of the QR Code–Based Web Evaluation System for Extension Office Projects is designed as a web-based platform that integrates users, application processes, and a centralized database to ensure efficient and seamless operations. It follows a client-server structure, allowing users to access the system through web browsers on their devices. By scanning a generated QR code, users are directed to specific evaluation forms or project records, enabling quick and convenient data submission. The architecture supports real-time data access, enhances workflow efficiency, and offers a reliable and user-friendly solution for managing and evaluating extension office projects. The system architecture of the QR Code Based Web Evaluation System for Extension Office Projects is structured to efficiently link users, administrators, and the database. At its core is a web-based platform that acts as the central point for all system activities and interactions. Beneficiaries or participants can conveniently access the system by scanning QR codes or submitting required information, enabling them to provide feedback and evaluations. Meanwhile, administrators handle tasks such as managing projects, generating QR codes, and reviewing reports, while staff members are responsible for examining responses and analyzing the gathered data. All relevant information including project details, QR codes, user data, and evaluation results is stored in a centralized database to ensure secure and well-organized data management. In addition, the system supports the extension office by delivering essential project information, evaluation outcomes, and analytical reports. Overall, this architecture promotes a seamless flow of information, allowing real-time monitoring, efficient data handling, and better decision-making.

Context Diagram

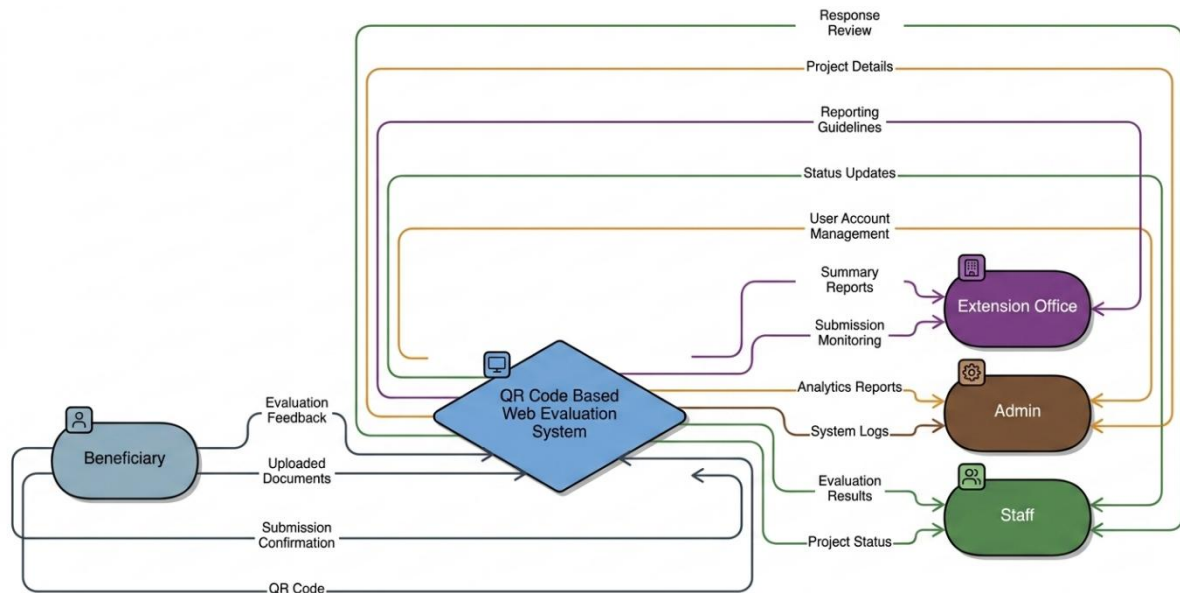


Figure 2-3. Context Diagram of the QR Code–Based Web Evaluation System

The diagram illustrates a simple but efficient flow of information where data is collected from beneficiaries, processed by the system, and then shared with staff, administrators, and the extension office. This digital approach improves accuracy, saves time, and makes evaluation more accessible and organized. The Context Diagram of the QR Code–Based Web Evaluation System for Extension Office Projects illustrates the overall interaction between the system and its external entities. At the center is the web-based evaluation system, which serves as the primary platform for collecting, processing, and managing evaluation data. The Student/Visitor (Event Participant) interacts with the system by scanning a QR code to access the evaluation form, submitting responses, and receiving confirmation after submission. The Extension Staff (Encoder/Field Worker) is responsible for generating and distributing QR codes, as well as providing necessary project information to facilitate participant access. The Project In-Charge (Evaluator) utilizes the system to request evaluation results, define evaluation periods, and receive summarized reports and analytics for

decision-making purposes. Meanwhile, the System Administrator manages user accounts, project configurations, and overall system settings, while also monitoring system logs and activities. Supporting these interactions is the Database (Cloud/Web Server), which stores evaluation data, QR code records, and other relevant information, and enables efficient data retrieval and analysis. Overall, the context diagram presents a high-level view of how data flows between users and the system, highlighting its role in streamlining the evaluation process through QR code technology and centralized data management.

Data Flow Diagram

Level 0 DFD

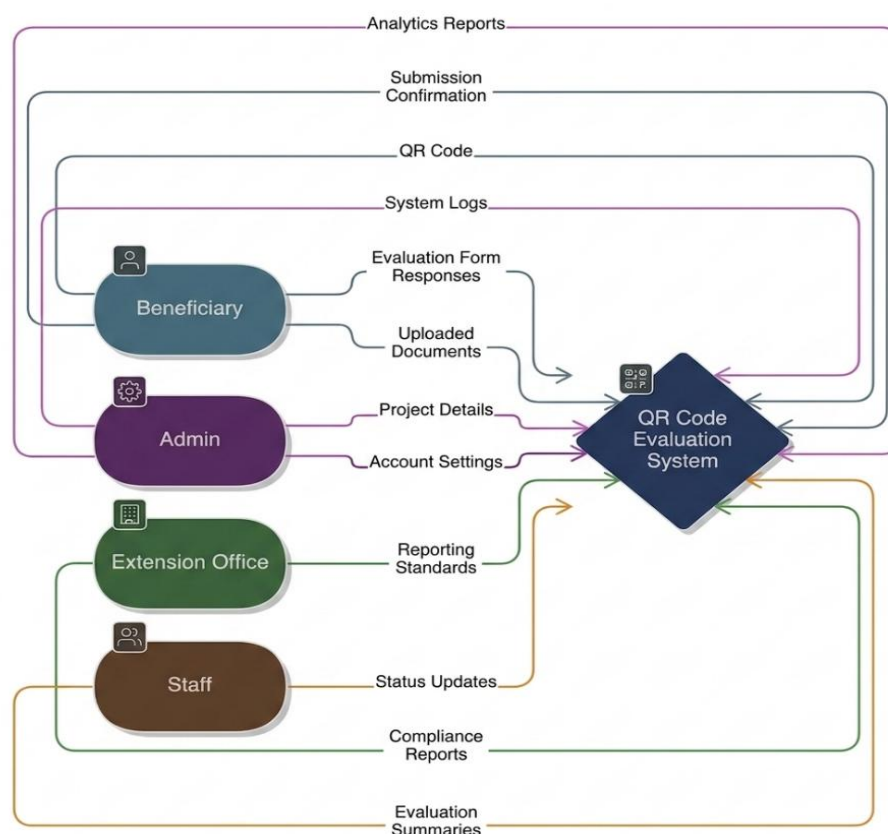


Figure 2-4. Level 0 Data Flow Diagram of the QR Code-Based Web Evaluation System

The diagram shows a Level 0 (context) view of a QR Code–Based Web Evaluation System, where different users interact with a central system. Students scan QR codes, access evaluation forms, and submit their responses, receiving confirmation after submission. Instructors access the system to view evaluation results, reports, and analytics. Administrators manage users, courses, and system settings while monitoring logs. All data—such as evaluations, user information, and QR codes—is stored and retrieved from a centralized database, ensuring smooth data flow and efficient evaluation processing.

Level 1 DFD

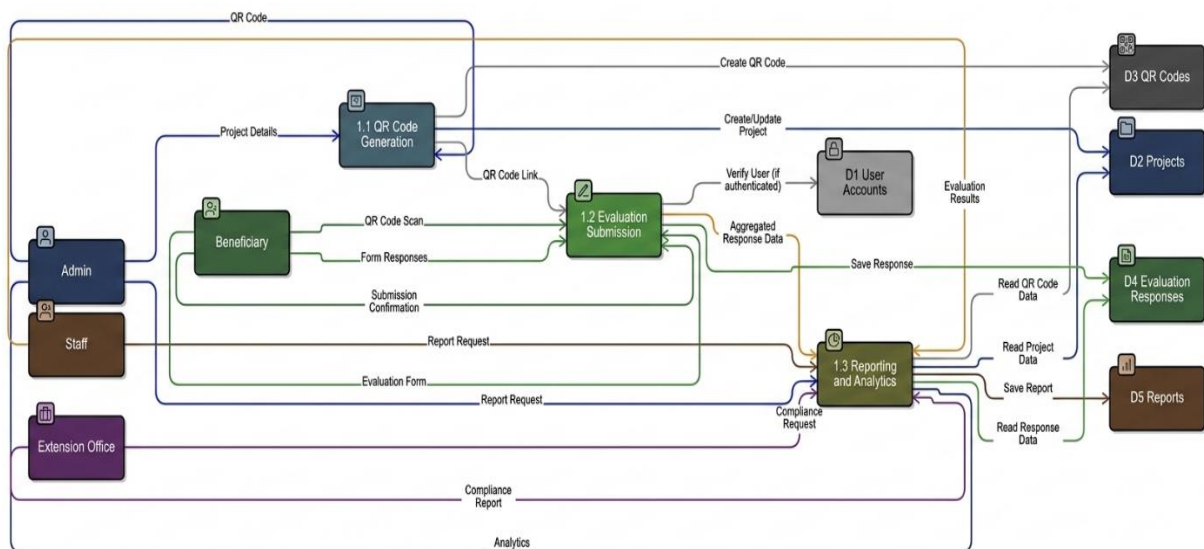


Figure 2-5. Level 1 Data Flow Diagram of the QR Code–Based Web Evaluation System

The figure above shows the Level 1 Data Flow Diagram of the QR Code–Based Web Evaluation System, such as the three key processes: (1.1) QR Code Generation, (1.2) Evaluation Submission, and (1.3) Reporting and Analytics. Furthermore, the figure also shows the external entities (Beneficiary, Admin, Staff, Extension Office) and data stores including D2 Projects, D3 QR Codes, D4

Evaluation Responses, and D5 Reports, as well as the data flows between processes, entities, and data stores.

Use Case Diagram

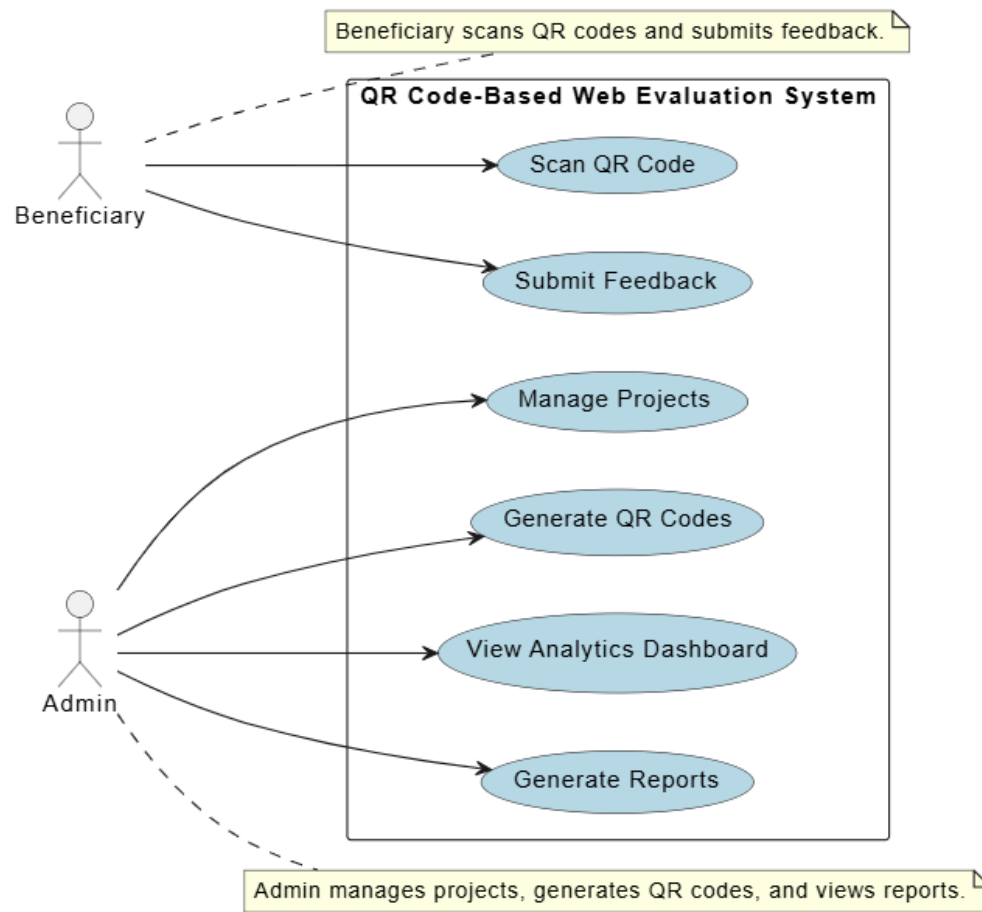


Figure 2-6. Use Case Diagram of the QR Code–Based Web Evaluation System

The figure above shows the use cases of the QR Code–Based Web Evaluation System through the use of a diagram. There are two types of users which have separate use cases in accessing the system which also have different privileges. Once a beneficiary scans the QR code, the system will display the evaluation form and the beneficiary can submit feedback. Once an admin logs into the system, the admin can manage projects, generate QR codes, view the analytics dashboard, and generate reports.

Entity Relationship Diagram (ERD)

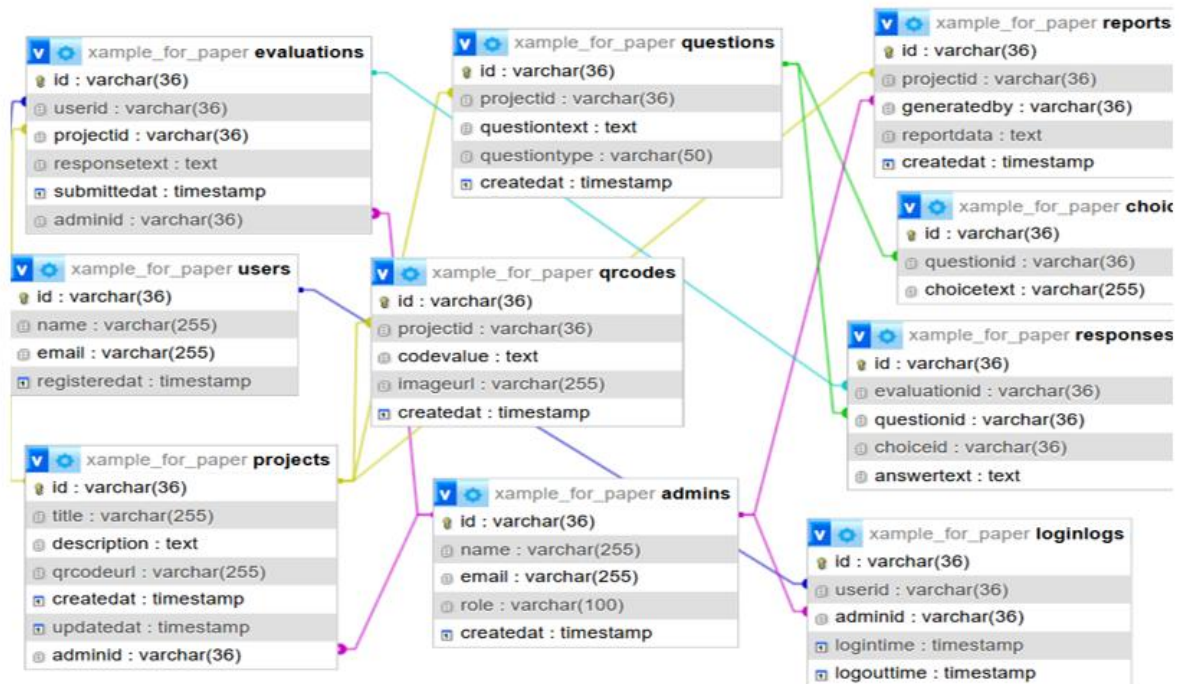


Figure 2-7. Entity Relationship Diagram of the QR Code–Based Web Evaluation System

The Entity Relationship Diagram (ERD) of the QR Code–Based Web Evaluation System presents the overall structure of the database by illustrating the key entities, their attributes, and the relationships among them.

The system is composed of four main entities; Users, Admins, Projects, and Evaluations. The Users entity contains information about individuals who participate in evaluating extension projects, while the admins entity stores the details of authorized personnel responsible for managing the system. The Projects entity represents the extension activities created and supervised by admins, each associated with a generated QR code that allows users to access the evaluation form. Meanwhile, the Evaluations entity serves as the central component of the system, as it records the feedback submitted by users for specific projects. Relationships between

these entities are established through foreign keys, where a single user can submit multiple evaluations, and each project can receive multiple evaluations, forming one-to-many relationships. Additionally, admins are linked to both projects and evaluations, indicating their role in creating projects and overseeing the evaluation process.

Database Schema

The diagram shows thirteen tables in a relational database for A QR Code-Based Web Evaluation System for Extension Office Projects System.

Table 2-1. Schema for users

Field Name	Data Type	Description	Constraints
Id	VARCHAR(36)	Unique identifier of the user	Primary Key
name	VARCHAR(255)	Full name of the user	NOT NULL
email	VARCHAR(255)	Email address of the user	NOT NULL, UNIQUE
registerdate	TIMESTAMP	Date and time of registration	DEFAULT CURRENT_TIMESTAMP

The Users table stores the information of individuals who participate in evaluating extension projects. These users access the system by scanning QR codes and submitting their feedback through the evaluation form.

Table 2-2. Schema for admin

Field Name	Data Type	Description	Constraints
Id	VARCHAR(36)	Unique identifier of the admin	Primary Key
name	VARCHAR(255)	Full name of the admin	NOT NULL
email	VARCHAR(255)	Email address of the admin	NOT NULL, UNIQUE
role	VARCHAR(100)	Role of the admin (e.g., staff)	NOT NULL
createdate	TIMESTAMP	Account creation date	DEFAULT CURRENT_TIMESTAMP

The Admin table contains the records of system administrators who manage the QR code-based evaluation system. They are responsible for creating projects, generating QR codes, and monitoring submitted evaluations

Table 2-3. Schema for projects

Field Name	Data Type	Description	Constraint
id	VARCHAR(36)	Unique identifier of the project	Primary Key
title	VARCHAR(255)	Title of the project	NOT NULL
description	TEXT	Description of the project	NULL
qrcodeurl	VARCHAR(255)	URL of the generated QR code	NULL
createdat	TIMESTAMP	Date the project was created	DEFAULT CURRENT_TIMESTAMP
updatedate	TIMESTAMP	Last update of the project	ON UPDATE CURRENT_TIMESTAMP
adminid	VARCHAR(36)	Admin who created the project	Foreign Key → admins(id)

The Projects table stores information about the extension projects that are subject to evaluation. Each project is created by an admin and is associated with a QR code that users can scan to access the evaluation form.

Table 2-4. Schema for evaluation

Field Name	Data Type	Description	Constraints
Id	VARCHAR(36)	Unique identifier of the evaluation	Primary Key
userid	VARCHAR(36)	User who submitted the evaluation	Foreign Key → users(id)
projectid	VARCHAR(36)	Project being evaluated	Foreign Key → projects(id)
responsetext	TEXT	Feedback or response of the user	NULL
submittedat	TIMESTAMP	Date and time of submission	DEFAULT CURRENT_TIMESTAMP
adminid	VARCHAR(36)	Admin overseeing the evaluation	Foreign Key → admins(id)

The Evaluations table serves as the central repository of all feedback submitted by users. It connects users, projects, and admins, allowing the system to track which user evaluated which project and when the evaluation was submitted.

Table 2-5. Schema for qrcodes

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier of the QR evaluation	Primary Key
projectid	VARCHAR(36)	Associated project	Foreign Key → projects(id)
codevalue	TEXT	Encoded QR code value or link	NOT NULL
imageurl	VARCHAR(255)	Stored QR code image path	NULL
createdat	TIMESTAMP	Admin overseeing the evaluation	DEFAULT CURRENT_TIMESTAMP

This table stores the generated QR codes for each project. Each QR code is linked to a specific project and contains a code value and image URL, allowing users to access evaluation forms by scanning it.

Table 2-6. Schema for questions

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier of the QR evaluation	Primary Key
projectid	VARCHAR(36)	Related project	Foreign Key → projects(id)
questiontext	TEXT	The evaluation question	NOT NULL
questiontype	VARCHAR(50)	The evaluation question	NOT NULL
createdat	TIMESTAMP	Date question was created	DEFAULT CURRENT_TIMESTAMP

This table contains the list of questions used in each project evaluation. It defines what users will answer and supports different question types such as multiple choice or text-based questions.

Table 2-7. Schema for choices

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier of the QR evaluation	Primary Key
questionid	VARCHAR(36)	Related question	Foreign Key → questions(id)
choicetext	VARCHAR(255)	Option text for multiple-choice answers	NOT NULL

This table stores the possible answer options for multiple-choice questions. It is linked to the questions table and helps standardize user responses for easier analysis.

Table 2-8. Schema for reports

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier of the report	Primary Key
projectid	VARCHAR(36)	Related project	Foreign Key → projects(id)
generatedby	VARCHAR(36)	Admin who generated the report	Foreign Key → admins(id), NULL
reportdata	TEXT	Stored evaluation summary/analysis	NULL
createdat	TIMESTAMP	Date report was generated	DEFAULT CURRENT_TIMESTAMP

This table stores the summarized results of evaluations. It is generated by administrators and contains analysis or reports based on user feedback for each project.

Table 2-9. Schema for loginlogs

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier of the log	Primary Key
userid	VARCHAR(36)	User who logged in (if user)	Foreign Key → users(id), NULL
adminid	VARCHAR(36)	Admin who logged in (if admin)	Foreign Key → admins(id), NULL
logintime	TIMESTAMP	Time user/admin logged in	DEFAULT CURRENT_TIMESTAMP
logouttime	TIMESTAMP	Time user/admin logged out	NULL

This table records all login and logout activities of users and administrators. It helps monitor system access and improves security by tracking who accessed the system and when.

GANTT CHART

The development timeline outlines the activities needed for researchers to complete the study and estimates the duration for each activity.

Table 2.18 Activities for March 2026

Month	March			
Activities	Week 1	Week 2	Week 3	Week 4
Record Office Staff Interviews				
Document Tracking Process Analysis				
Storage & Retrieval Process Analysis				
Document Submission Process Analysis				

Table 2.19 Activities for April 2026

Month	April			
Activities	Week 1	Week 2	Week 3	Week 4
User Account Management Design				
QR Code Generator Module Design				
Evaluation Form Interface Design (Mobile-Responsive)				
Admin Dashboard Design				

Table 2.20 Activities for May 2026

Month	May			
Activities	Week 1	Week 2	Week 3	Week 4
Development: User Account Module				
Development: QR Code Generator Module				
Development: Evaluation Form Interface				
Data Validation Integration				

Table 2.21 Activities for June 2026

Month	June			
Activities	Week 1	Week 2	Week 3	Week 4
Development: Admin Dashboard				
Development: Reporting & Analytics Module				
Integration of All Modules				
Initial System Testing				

Table 2.22 Activities for August 2026

Month	August			
Activities	Week 1	Week 2	Week 3	Week 4
System Condition Check				
Alpha Testing				
Bug Fixing & Adjustments				

Table 2.23 Activities for September 2026

Month	September			
Activities	Week 1	Week 2	Week 3	Week 4
Beta Testing Preparation				
Deployment for Selected Users				
Feedback Collection				

Table 2.24 Activities for October 2026

Month	October			
Activities	Week 1	Week 2	Week 3	Week 4
Beta Testing Execution				
System Refinements				
Performance Optimization				

Table 2.25 Activities for November 2026

Month	November			
Activities	Week 1	Week 2	Week 3	Week 4
System Evaluation (ISO/IEC 25010)				
Data Analysis & Interpretation				
Final System Documentation				

Table 2.26 Activities for December 2026

Month	December			
Activities	Week 1	Week 2	Week 3	Week 4
Final System Documentation				
System Deployment				
Final Presentation & Submission				

CHAPTER III

METHODOLOGY

Software Development Methodology

The study adopts the Agile Software Development Methodology, specifically the Scrum framework, due to its iterative and adaptive nature. Scrum allows for continuous improvement through short development cycles called sprints, each lasting one to two weeks. This approach is ideal for the project because it enables frequent feedback from Extension Office personnel and project beneficiaries, ensuring that the final system closely matches actual needs.

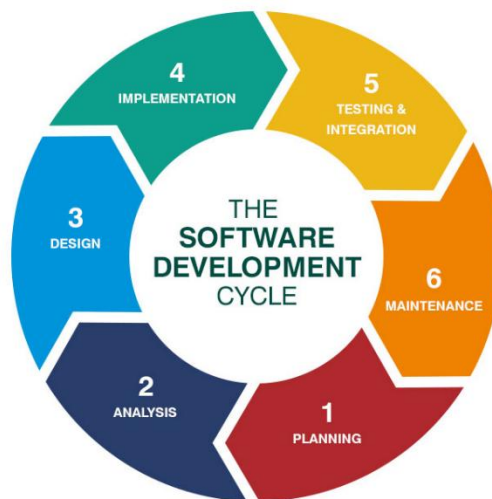


Figure 3-1. Scrum Framework (Agile Software Development Methodology) of the QR Code–Based Web Evaluation System

Phase 1: Product Backlog Creation

The researchers conduct interviews with the Extension Office head, extension coordinators, and project leaders using a semi-structured questionnaire to identify problems with the current paper-based evaluation process. Common issues such as misplaced forms, delayed data consolidation, and lack of real-time reporting are

documented. These findings are used to define and prioritize system features in the Product Backlog, including user authentication, QR code generation, mobile-responsive forms, real-time submission, admin dashboard, automated validation, and response analytics.

Phase 2: Sprint Planning

The researchers translate the prioritized Product Backlog into time-boxed sprints. Each sprint begins with a planning meeting to define sprint goals and deliverables. Tasks are assigned and estimated in hours or story points. The sprints are organized as follows: Sprint 0 (project setup and infrastructure), Sprint 1 (user authentication and project management), Sprint 2 (QR code generation and linking), Sprint 3 (evaluation form system), Sprint 4 (admin dashboard and reporting), and Sprint 5 (testing, refinement, and pilot deployment).

Phase 3: Sprint Execution (Development)

The researchers design, code, and test selected features in each sprint using a three-tier architecture. For front-end development, HTML5, CSS3, Bootstrap 5, and JavaScript are used to create a mobile-first responsive interface. For back-end development, PHP handles server-side logic including authentication, form processing, QR code generation, and reporting, using the XAMPP server environment with Apache and MySQL. A QR code generation library is integrated to create unique, project-specific static QR codes containing URLs to evaluation form

System Wireframe

The login form is centered on a light green background. It features a QR code icon at the top, followed by input fields for Username and Password. A green Login button is positioned below the password field, with a [Forgot Password?](#) link underneath. The Isabela State University logo and tagline are at the bottom of the form. The footer of the page contains the copyright notice: © 2023 Isabela State University - ICT Department. All Rights Reserved.

ADMIN & STAFF LOGIN

Username
Enter your Username

Password
Enter your Password

Login

[Forgot Password?](#)

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Figure 3-2. Login Form



Figure 3-3. Admin Dashboard

Create Project

Project Title:

Project Description:

Start Date: End Date:

Generate QR Code

QR Code Preview [Download](#)

Figure 3-4. Create Project Page

Projects

Search: [Search](#)

Filter: **All** (Active, Completed)

Project Title	QR Code	Date Created	Status	Actions
QR Code Project 1		27 May 2023	Active	View Edit Delete Generate QR
QR Code Project 2		27 May 2023	Active	View Edit Delete Generate QR
QR Code Project 3		27 May 2023	Completed	View Edit Delete Generate QR
QR Code Project 4		26 May 2023	Completed	View Edit Delete Generate QR

1 2 3 4 5 6 ... 8 Next

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Figure 3-5. Project List Page

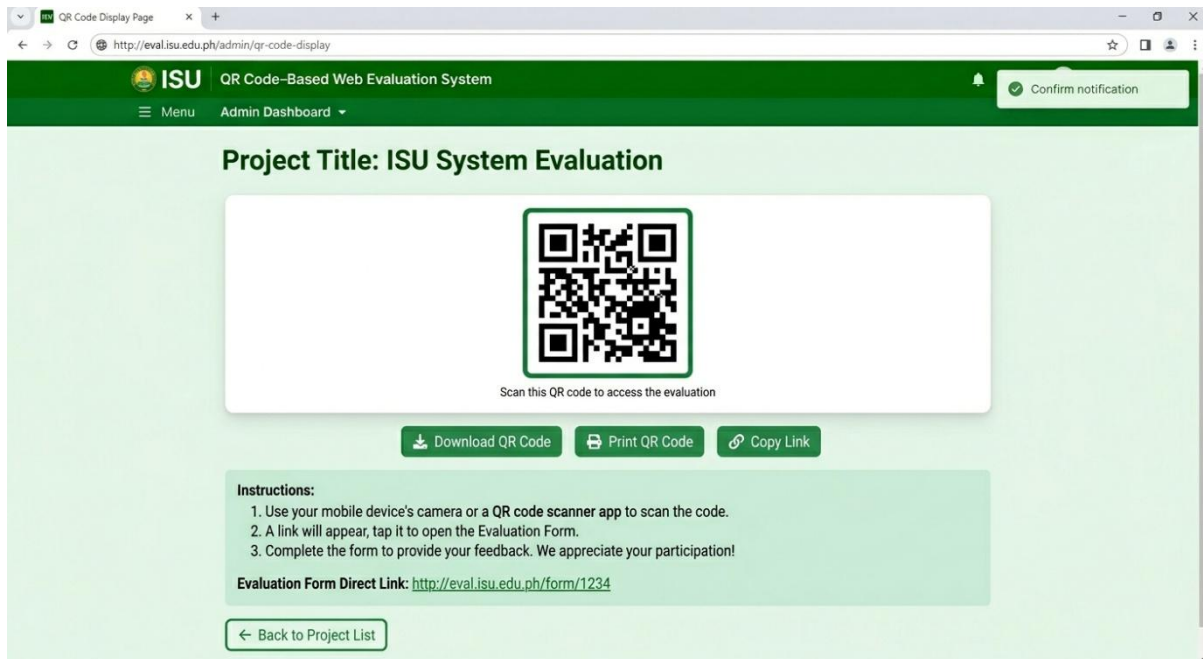


Figure 3-6. QR Code Display Page

ISU Evaluation: ISU QR Code Project 1

40% Complete

Q1: Please rate the clarity of the presentation.

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q2: Please rate the ctality of the presentation.

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Q3: Please rate the community pesciations.

★ 1 ★ 2 ★ 3 ★ 4 ★ 5

Comment optional (optional)

SUBMIT EVALUATION

Please answer all required questions.
For issues, contact: [ICT Dept. Support](#)

Figure 3-7. Evaluation Form

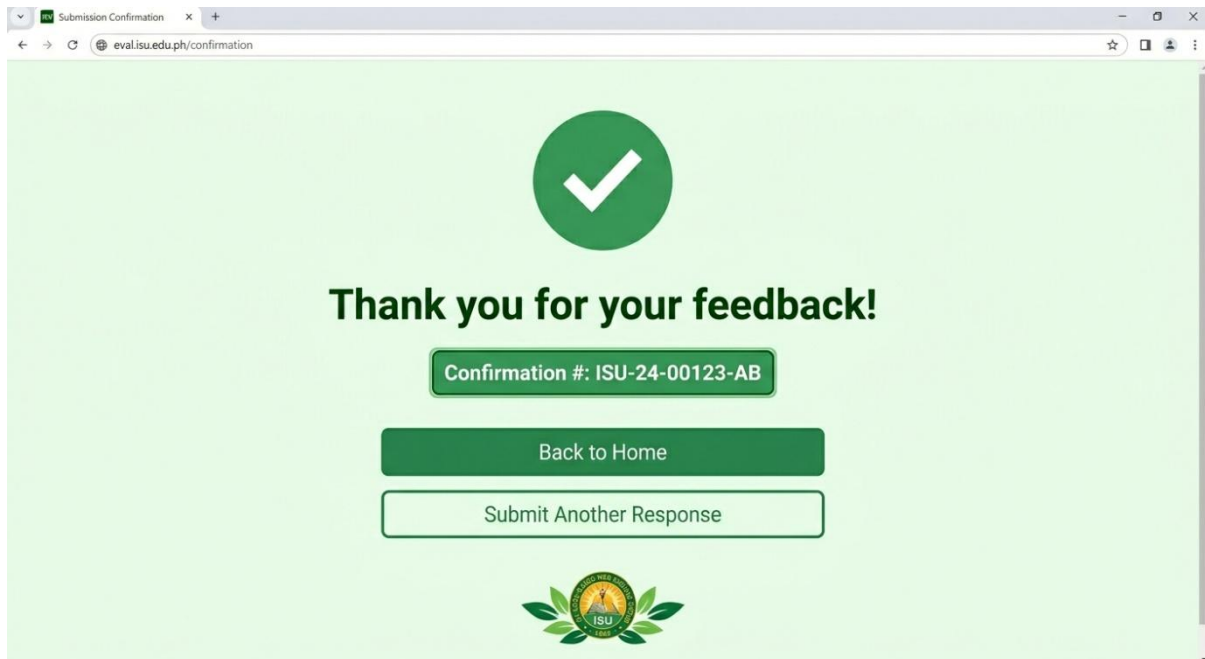


Figure 3-8. Submission Confirmation Page

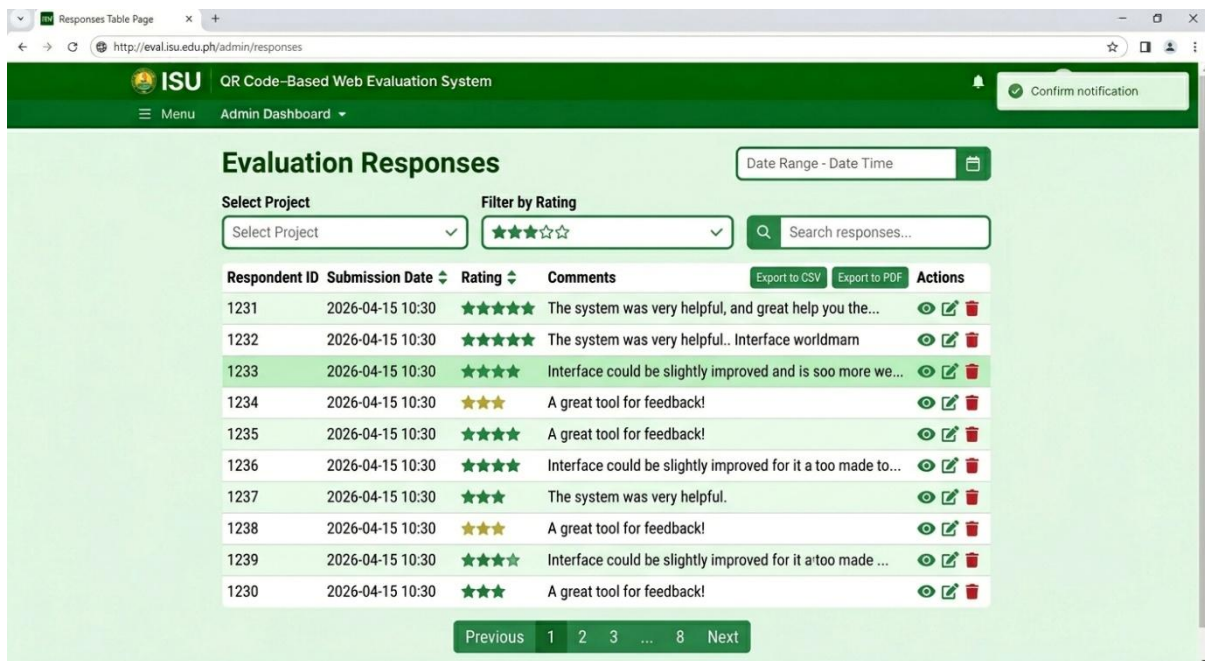


Figure 3-9. Responses Table Page



Figure 3-10. Reports and Analytics Page

Phase 4: Daily Scrum Meetings

During each sprint, the researchers conduct short daily meetings to monitor progress, identify blockers, and ensure collaboration. Each meeting addresses three questions: what was accomplished yesterday, what will be done today, and what issues are encountered.

Phase 5: Sprint Review

At the end of each sprint, the researchers conduct a Sprint Review meeting with the Extension Office personnel and capstone adviser. The working system increment is demonstrated, and stakeholders provide feedback on functionality and usability. Suggested changes are documented and added to the Product Backlog for future sprints.

Phase 6: Sprint Retrospective

After each Sprint Review, the researchers conduct a Sprint Retrospective meeting focused on the development process. The team evaluates what went well, what did not go well, and what can be improved in the next sprint to continuously enhance workflow efficiency.

Phase 7: Incremental System Development

The system is developed incrementally, with features added gradually over multiple sprints. After Sprint 1, a basic system with login and project management is available. After Sprint 2, QR code generation is added. After Sprint 3, the full evaluation form system is operational. After Sprint 4, the admin dashboard and reporting features are available. After Sprint 5, a fully tested and refined system is ready for pilot deployment.

Phase 8: Testing and Continuous Integration

Testing is performed continuously during each sprint. Unit testing, integration testing, functional testing, usability testing, security testing, and compatibility testing are conducted. Bugs or issues identified are added to the Product Backlog and prioritized for resolution in the current or next sprint.

Phase 9: Final Deployment and Evaluation

After all sprints are completed, the system is deployed for pilot testing with actual extension projects. Extension staff receive a brief orientation on creating projects, generating QR codes, and accessing the admin dashboard. The system is then used for three to five actual extension projects, with beneficiaries scanning QR

codes and submitting evaluation responses. The final evaluation uses ISO/IEC 25010 software quality standards and TAM/UTAUT user acceptance questionnaires.

Respondents of the Study

Respondents	Frequency	Percentage
Employees	13	65%
IT Professionals	5	25%
End Users	2	10%
Total	20	100%

This table presents the distribution of respondents involved in the system evaluation. The majority are employees, including faculty members and administrative staff, while the remaining respondents are IT professionals. This distribution ensures a balanced evaluation, incorporating both user experience and technical perspectives, which is essential for assessing the system's usability, functionality, and overall performance.

Data Gathering Procedure

The data gathering procedure is a systematic process used to collect relevant information necessary for the study. First, the researchers prepared to obtain permission to conduct the study. After approval, coordination with the office personnel was established.

Second, interviews were conducted with Extension Office staff to understand the existing evaluation process, identify challenges, and gather system requirements. Third, direct observation was carried out to examine how evaluation forms are distributed, answered, collected, and stored. This helped identify inefficiencies in the current system. Fourth, a survey questionnaire was distributed to selected respondents. The questionnaire used a Likert scale to measure user satisfaction, usability, and acceptance of the proposed system. Fifth, document analysis was performed by reviewing existing evaluation forms, reports, and records to validate the

gathered data. Finally, all collected data were organized, analyzed, and used as the basis for designing and developing the proposed system.

Sampling Technique

The study will employ purposive sampling for selecting Extension Office personnel, project leaders, and IT experts, as these individuals will be selected based on specific criteria related to their expertise and direct involvement with extension project evaluation. All 8 extension personnel and all 5 IT experts will be included in the study (total of 13) due to their small population size.

For the project participants and community beneficiaries, the study will employ simple random sampling to ensure that every beneficiary has an equal chance of being selected. The sample size will be determined using Slovin's Formula:

$$n = \frac{N}{1 + Ne^2}$$

Where:

- n = required sample size
- N = total population size
- e = margin of error (set at 5% or 0.05)

Calculation:

$$n = \frac{320}{1 + 320(0.05)^2}$$

$$n = \frac{320}{1 + 320(0.0025)}$$

$$n = \frac{320}{1 + 0.8}$$

$$n = \frac{320}{1.8}$$

$$n = 177.78 \approx 178$$

Thus, the required sample size for project participants and community beneficiaries will be 178. This sample size will provide a 95% confidence level with a 5% margin of error.

Statistical Treatment

The following statistical treatments will be used to analyze the data collected from the respondents. These methods will ensure that the evaluation of the QR Code–Based Web Evaluation System is systematic, objective, and reliable.

1. Frequency and Percentage

Frequency and percentage will be used to determine the distribution of respondent profiles, such as the number of extension personnel, project participants, and IT experts. This analysis will help describe the composition of the sample. The formula is:

$$\% = \left(\frac{F}{N} \right) \times 100$$

Where:

- % = percentage
- F = frequency (number of responses in a category)
- N = total number of respondents

2. Weighted Mean

Weighted mean will be used to calculate the average rating for each indicator in the ISO/IEC 25010 software quality questionnaire and the TAM/UTAUT user acceptance questionnaire. The formula is:

$$WM = \frac{\sum fx}{N}$$

Where:

- WM = weighted mean
- $\sum fx$ = sum of the product of frequency (f) and weight (x)
- N = total number of respondents

3. Likert Scale Interpretation

A four-point Likert scale will be used to interpret the responses from all questionnaires. The four-point scale eliminates a neutral option, forcing respondents to express a clear opinion. The scale ranges and verbal interpretations are as follows:

Numerical Scale	Range	Interpretation
5	4.21 - 5.00	Strongly Agree
4	3.41 - 4.20	Agree
3	2.61 - 3.40	Neutral
2	1.81 - 2.60	Disagree
1	1.00 - 1.80	Strongly Disagree

4. Cronbach's Alpha

Cronbach's alpha will be used to test the internal consistency and reliability of the survey questionnaires. This coefficient measures how closely related a set of items are as a group. The formula is:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2} \right)$$

Where:

- α = Cronbach's alpha coefficient
- k = number of items in the questionnaire
- $\sum \sigma_i^2$ = sum of the variances of each item
- σ_t^2 = variance of the total scores

A Cronbach's alpha value of 0.70 or higher will be considered acceptable, indicating that the questionnaire has good internal consistency.

5. ISO/IEC 25010 Interpretation

The ISO/IEC 25010 software quality model will be used to evaluate the system across six quality characteristics: functional suitability, reliability, usability, performance efficiency, security, and maintainability. For each characteristic, the weighted mean will be computed using the formula:

$$\text{Mean Score per Characteristic} = \frac{\sum(\text{respondent ratings})}{\text{number of respondents}}$$

The resulting mean score for each quality characteristic will be interpreted using the four-point Likert scale described above. A mean score of 3.26 to 4.00 will indicate that the system fully meets the quality standard for that characteristic, while a score below 2.51 will indicate areas requiring improvement.

6. TAM and UTAUT Evaluation

The Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) will be used to measure user acceptance of the system. The following constructs will be evaluated:

Framework	Construct	Description
TAM	Perceived Usefulness	Degree to which users believe the system improves their evaluation tasks
TAM	Perceived Ease of Use	Degree to which users believe the system is easy to use
UTAUT	Performance Expectancy	Belief that using the system will enhance job performance
UTAUT	Effort Expectancy	Ease of using the system
UTAUT	Social Influence	Degree to which users perceive that others believe they should use the system
UTAUT	Facilitating Conditions	Degree to which users believe organizational and technical infrastructure supports system use
Both	Behavioral Intention to Use	Users' intention to continue using the system

For each construct, the weighted mean will be computed using the same formula as in ISO/IEC 25010:

$$\text{Mean Score per Construct} = \frac{\sum(\text{respondent ratings})}{\text{number of respondents}}$$

The resulting mean scores will be interpreted using the four-point Likert scale. A mean score of 3.26 to 4.00 will indicate high user acceptance, while scores below 2.51 will indicate low acceptance and the need for system refinements.

Technical Evaluation

In addition to the ISO/IEC 25010 software quality evaluation and TAM/UTAUT user acceptance assessment, a dedicated technical evaluation will be conducted to measure the system's performance, reliability, and compatibility under various conditions. This evaluation will provide objective, quantifiable data on how the system behaves in real-world usage scenarios.

Technical Performance Metrics

The following seven metrics will be measured during the technical evaluation:

Metric	Description	Target
Response Time	Time (in seconds) from QR code scan to full loading of evaluation form	≤ 3 seconds
QR Code Scanning Success Rate	Percentage of QR codes that correctly redirect to the intended evaluation form	100%
Report Generation Time	Time (in seconds) to generate a summary report for a project with 100 responses	≤ 5 seconds
Data Submission Success Rate	Percentage of completed forms successfully saved to the database	$\geq 99\%$
Browser Compatibility	Number of browsers on which all system functions work correctly	5/5 browsers
Mobile Responsiveness	Percentage of interface elements that display correctly on mobile screens	$\geq 95\%$
System Performance Under Multiple Users	Maximum number of concurrent users before response time exceeds 5 seconds	≥ 100 users

Technical Test Procedures

The following five technical tests will be conducted to evaluate the system.

1. QR Code Scan Test

Twenty randomly generated QR codes for different extension projects will be tested. Each QR code will be scanned using five different smartphones (different brands and operating systems). The test will verify that every QR code correctly redirects the user's browser to the intended project-specific evaluation form without errors or incorrect redirections. A success rate of 100% will be considered passing.

2. Response Time Test

The time required to load the evaluation form after scanning a QR code will be measured using browser developer tools and network throttling simulations. Measurements will be taken under three network conditions: high-speed Wi-Fi (≥ 10 Mbps), standard 4G mobile (5 Mbps), and slow 3G (1 Mbps). Ten measurements will be taken for each condition, and the average load time will be calculated. The target is a load time of three seconds or less under standard 4G conditions.

3. Compatibility Test

The system will be tested on five web browsers: Google Chrome (desktop and mobile), Mozilla Firefox (desktop), Microsoft Edge (desktop), Safari (iOS), and the default Android browser (Chrome for Android). For each browser, all system functions will be tested, including user login, project creation, QR code generation, form submission, admin dashboard access, and report generation. Any browser-specific display or functionality issues will be documented and categorized as critical, major, or minor.

4. Concurrent User Test

To simulate real-world usage where multiple beneficiaries may submit evaluations simultaneously after an extension project, the system will be tested using Apache JMeter or a similar load testing tool. Virtual users will be ramped up from 10 to 200 concurrent users, and response times and error rates will be recorded at each interval. The test will determine the maximum number of concurrent users the system can handle before response times exceed five seconds or error rates exceed 1%.

5. Security Test

The following security vulnerabilities will be tested:

- Invalid input: Attempt to submit form data with SQL injection patterns, cross-site scripting (XSS) payloads, and excessively long text entries.
- Duplicate submissions: Attempt to submit the same evaluation response multiple times within a short period.
- Unauthorized access: Attempt to access the admin dashboard, view evaluation responses, or generate reports without proper login credentials or session tokens.
- QR code manipulation: Attempt to modify the QR code URL to access evaluation forms for other projects without authorization.

Any security vulnerabilities identified will be documented and fixed before final deployment.

Comparison of Manual and Proposed System

Table 1 presents a direct comparison between the existing manual evaluation process and the proposed QR Code–Based Web Evaluation System across five criteria.

Table 1

Criterion	Existing Manual Process	Proposed System
Time to collect evaluation forms	Several days	Immediate
Time to consolidate data	1–2 weeks	Real time
Risk of missing forms	High	Very low
Cost of printing	High	Minimal
Accuracy of reports	Moderate	High

As shown in Table 1, the proposed system offers significant improvements over the manual process. Evaluation forms are collected immediately upon beneficiary submission rather than after several days of distribution and retrieval. Data consolidation occurs in real time, eliminating the 1–2-week delay required for manual encoding. The risk of misplaced or lost paper forms is reduced from high to very low through digital storage. Printing costs are minimized as forms are accessed digitally via QR codes. Finally, report accuracy is improved through automated data validation and real-time aggregation, reducing human encoding errors.

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